Faculty of Medicine Annaba
physiology course 1st year
medicine

Synaptic Transmission

neuromuscular Junction
neuro-neuronal synapses

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Plan

I/ Introduction

II/ Classifications of synapses

III/ Structure and general diagram of the functioning of a chemical synapse

IV/ Properties of chemical synapses

V/ Criteria for identifying a neurotransmitter and the different neurotransmitters

VI/ The neuromuscular junction (motor endplate)

VII/ neuro-neuronal Synapses

I/Introduction

The term synapse was proposed by Sherrington to designate areas of contiguity between neurons, specialized in the transmission of information.

The term synapse does not only refer to connections Between neurons (inter-neuronal synapses)

But also those between neurons and effector cells such as muscle and glandular cells (neuro-effectors synapses)

It is at the level of these contacts that the transmission of Information from one cell to another: synaptic transmission.

II/Classification of synapses

Synapses can be classified according to:

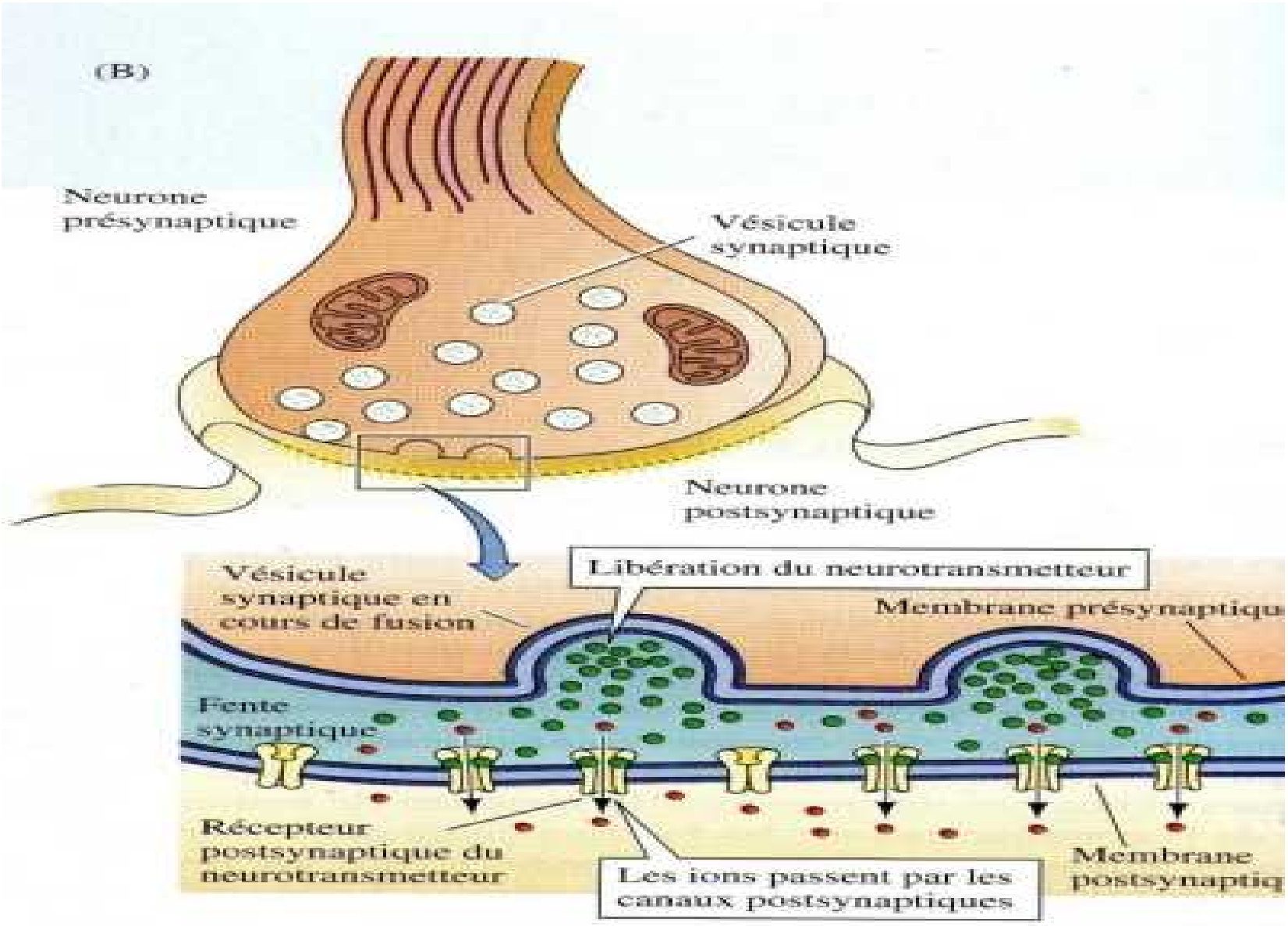
1. *the mode of operation*
2. *the nature of the postsynaptic element*
3. *the effect on the postsynaptic membrane*
4. *the nature of the chemical mediator*

1/DEPENDING ON THE OPERATING MODE

1-<u>CHEMICAL SYNAPSES</u>:

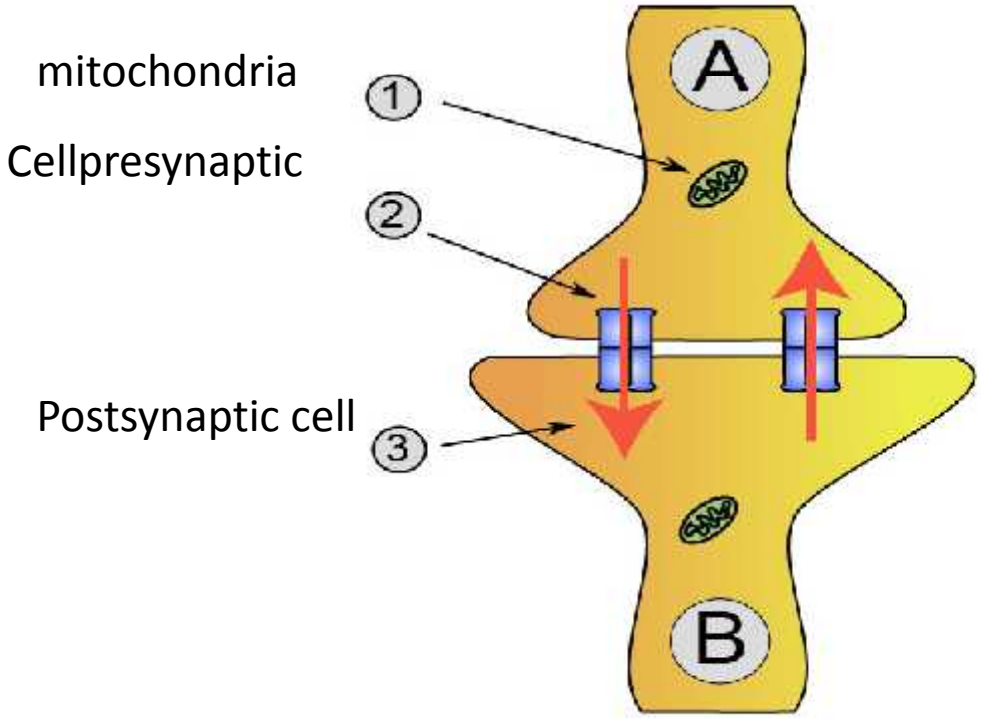
- *A **chemical substance** (**neurotransmitter**) is released by the nerve fiber presynaptic For **to settle down** on **receptors** located at the membrane level postsynaptic For **modify its excitability.***
- *The transmission of information is **<u>one-way</u>,** it is done from the presynaptic element to the postsynaptic element.*

Chemical synapse

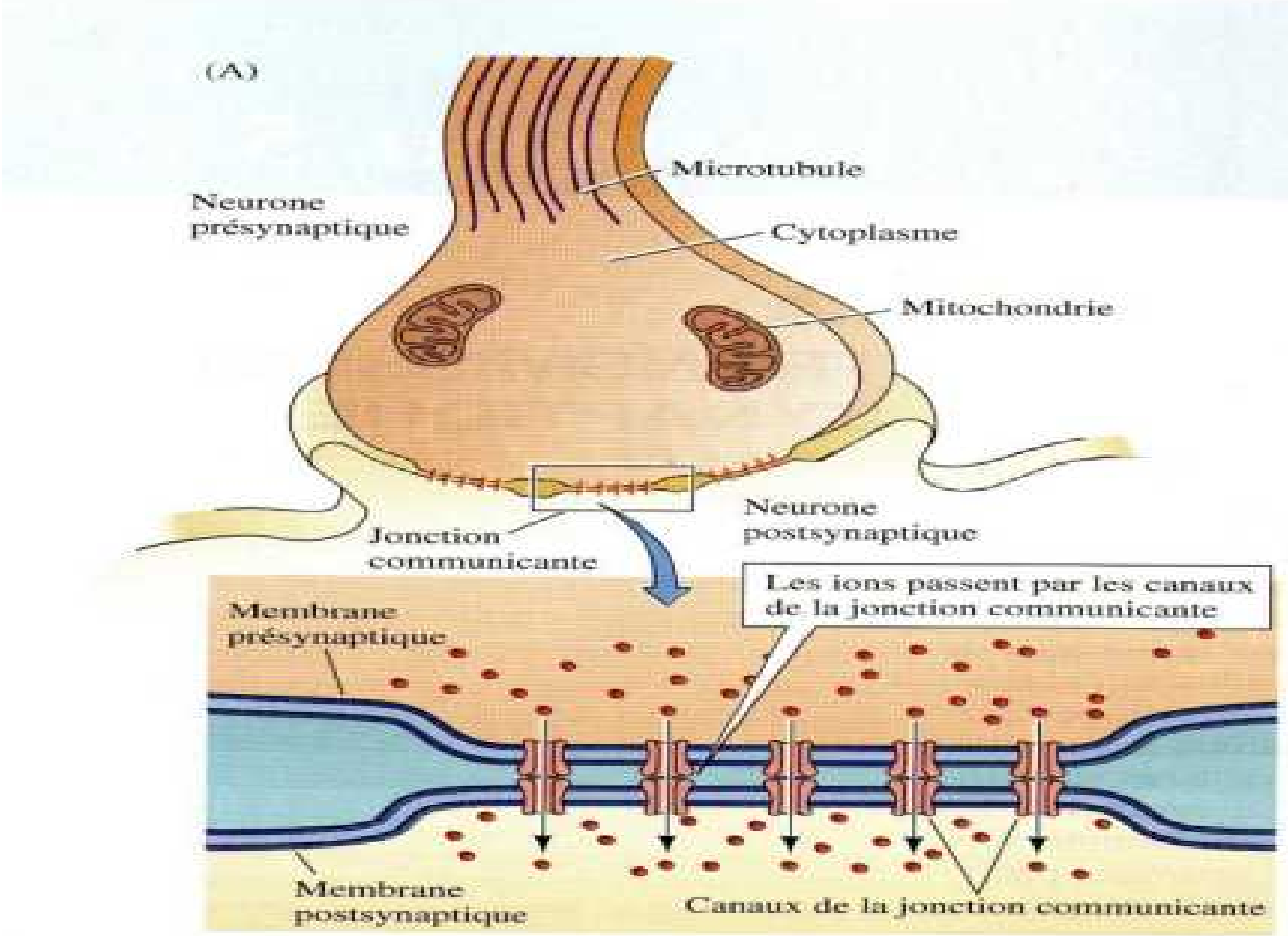


***2-<u>ELECTRICAL SYNAPSES</u>**: the interactions are of purely natural **electric** (variations in potential of a neuron's membrane are transmitted directly to another neuron).*

*The transmission is **<u>bidirectional.</u>***



Electrical synapse

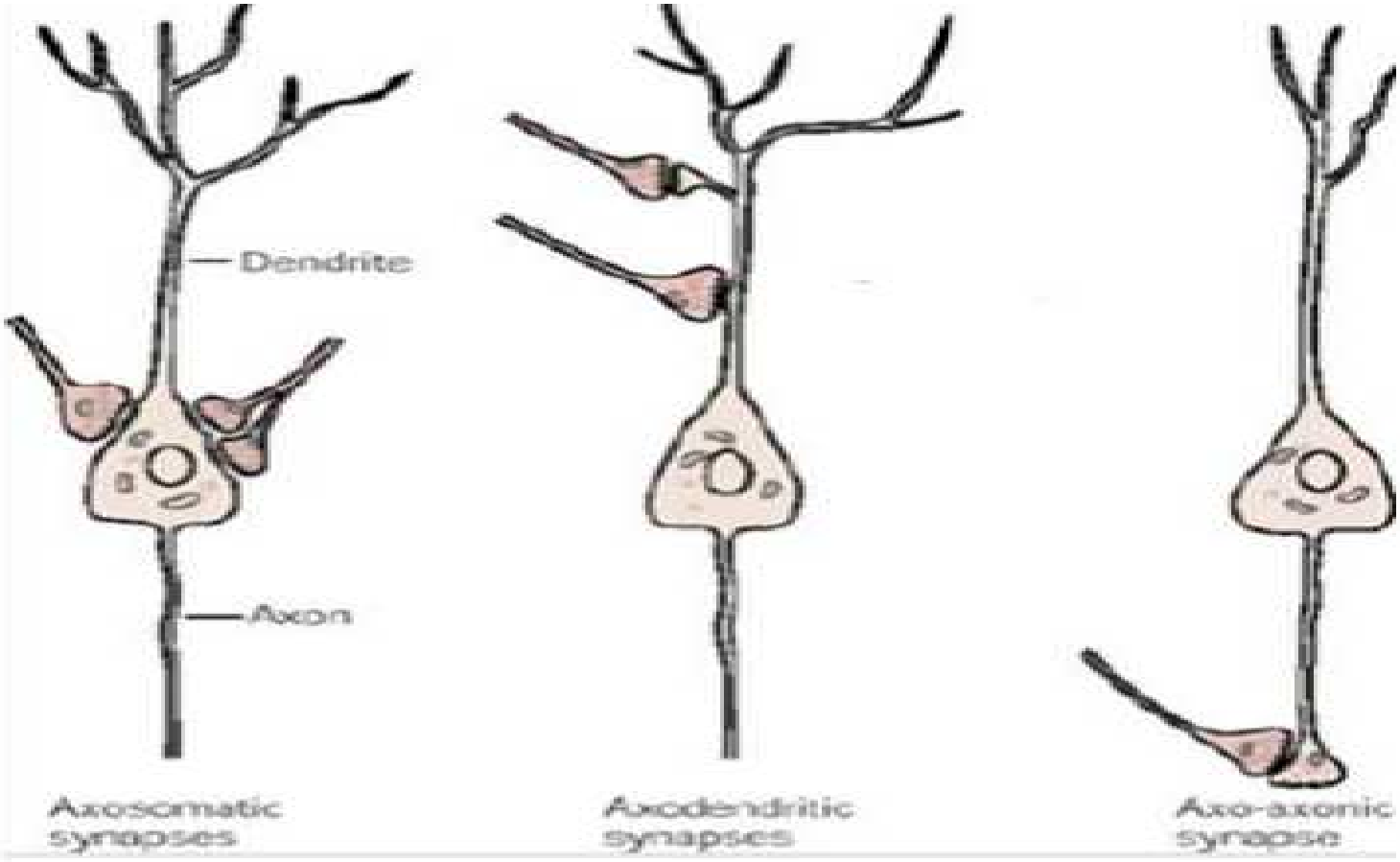


B/ ACCORDING TO THE NATURE OF THE POST-SYNAPTIC ELEMENT

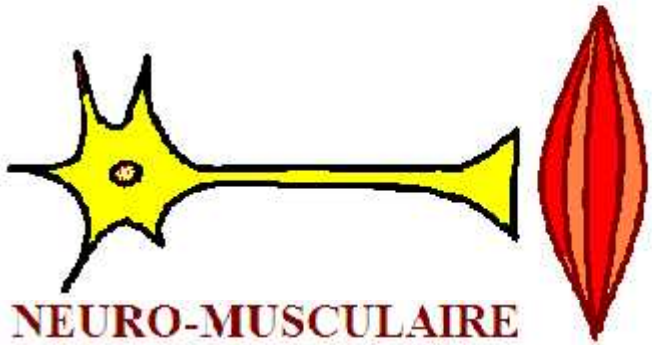
*1-**NEURO-NEURONAL SYNAPSE:**The element postsynaptic is a **nerve membrane**. (Who can be axo-somatic ,axo-dendritic ,axo-axonic).*

*2-**NEURO-EFFECTOR SYNAPSE:**The element postsynaptic is a membrane of a **cell effector** : striated skeletal or cardiac muscle – smooth muscle – glandular.*

Neuroneuronal Synapses



CATEGORIES DE SYNAPSES



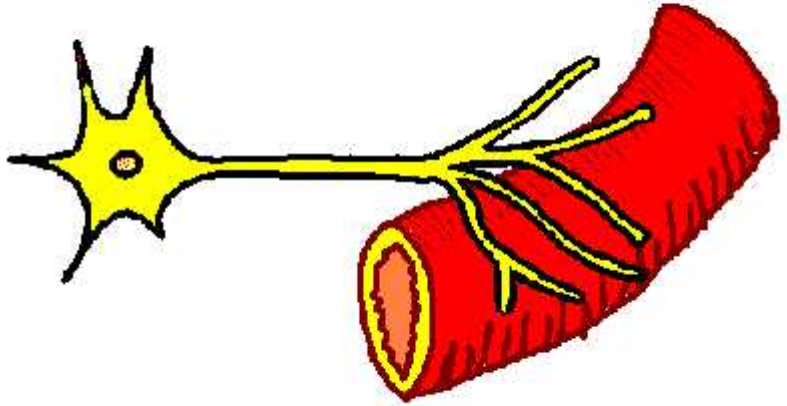
NEURO-MUSCULAIRE



NEURO-GLANDULAIRE



NEURO-NEURONALE



TERMINAISONS LIBRES

C/ACCORDING TO THE EFFECT ON THE POSTSYNAPTIC MEMBRANE

*1-**EXCITATORY SYNAPSES:** when activating the presynaptic element induces a **depolarization of the postsynaptic membrane.***

*2-**INHIBITORY SYNAPSES:**when activating the presynaptic element induces a **hyperpolarization***

D/ ACCORDING TO THE NATURE OF THE CHEMICAL MEDIATOR RELEASED

- *Cholinergic synapses (acetylcholine)*
- *Glutamatergic Synapses(glutamate)*
- *Gabaergic Synapses (GABA): Gamma amino butyric acid*
- *Dopaminergic synapses (dopamine)*
- *Peptidergic Synapses (peptides)*……

III/ STRUCTURE AND GENERAL DIAGRAM OF THE OPERATION OF A CHEMICAL SYNAPSE:

the synaptic complex is made up of 3 parts:
An presynaptic element (which is still a nerve cell),
An postsynaptic element (which can be either a nerve cell or an effector)
and A synaptic space or synaptic cleft.

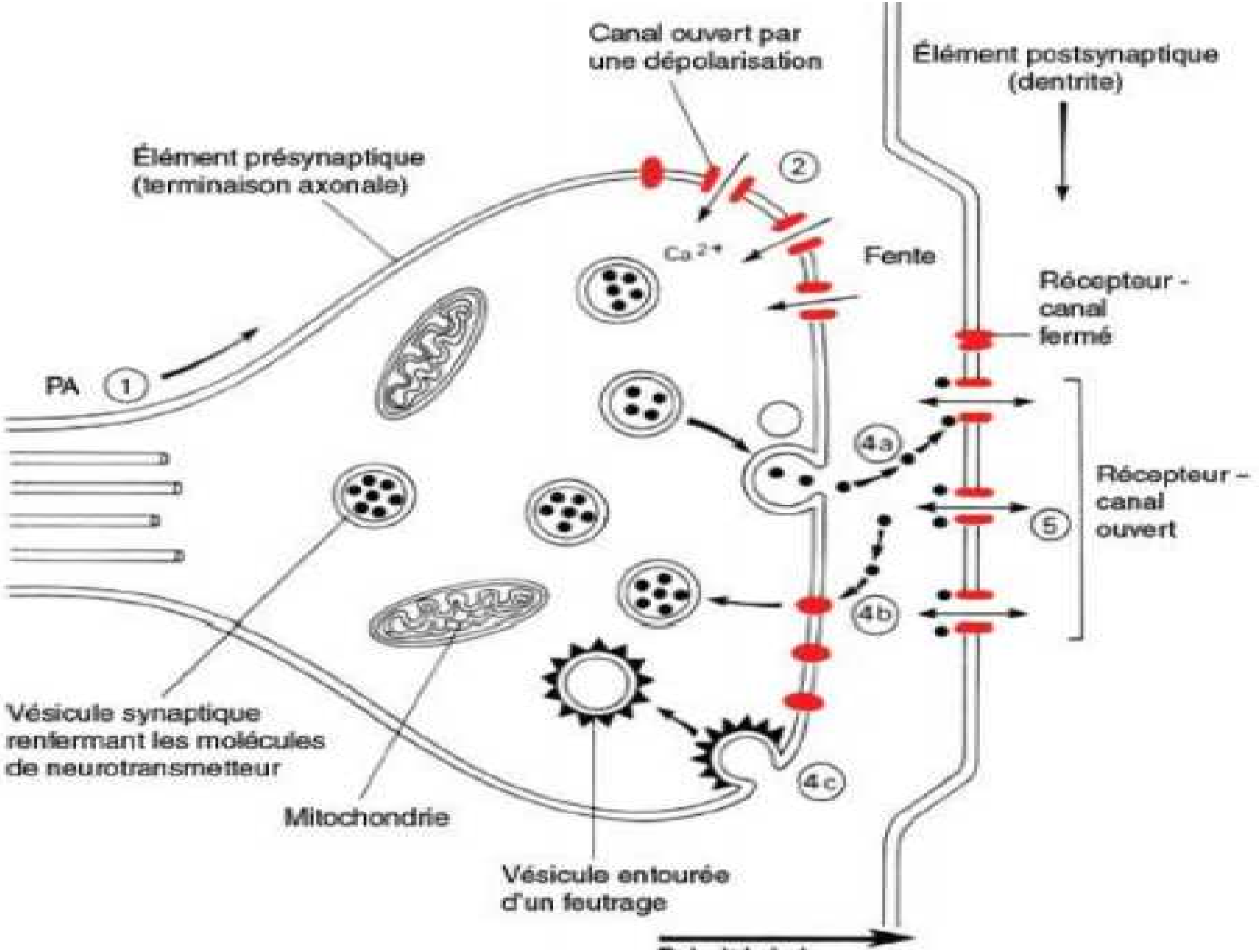
1/ The arrival of the Action Potential in the pre-synaptic – entry of Calcium

2/ The calcium signal induces the process of exocytosis by membrane fusion

3/ Release of vesicular contents into the synaptic cleft

4/ Binding of the neurotransmitter to a specific receptor

5/ Rapid inactivation by degradation, recapture, or by broadcast.


Canal ouvert par une dépolarisation
Élément postsynaptique (dentrite)
Élément présynaptique (terminaison axonale)
2
Ca 2+
Fente
Récepteur - canal fermé
PA
1
4a
Récepteur – canal ouvert
5
4b
Vésicule synaptique renfermant les molécules de neurotransmetteur
Mitochondrie
4c
Vésicule entourée d'un feutrage

IV-Properties of chemical synapses:

a/ Unidirectional: *the nerve impulse passes only from the presynaptic element towards the postsynaptic element*

b/ Fatigue: *repeated stimulations lead to a decrease progressive response in postsynaptic by exhaustion of the stock of neurotransmitters.*

c/ Synaptic delay*: time needed to release the neurotransmitter in the synaptic cleft and its action on the receiver of the post synaptic element*

d/ Integrative role: *the neuron in contact with several neurons is assimilated to an integrator the processed information can be modified*

e/ Does not obey the law of all or nothing

V/ Criteria for identifying a neurotransmitter

A neurotransmitter (or neuromediator) is a substance chemical released by a neuron at a synapse.
For a molecule to be considered a neurotransmitter, it must meet the following criteria:

- *be synthesized by the neuron;*
- *be present in the presynaptic termination ;*
- *be released in sufficient quantity to exert an action on the postsynaptic neuron ;*
- *reproduce the effect of the endogenous neurotransmitter when applies it directly to the target neuron;*
- *be specifically inactivated*

The different neurotransmitters:

we can subdivide them into 2 large groups:

- ***Classic neurotransmitters:***
  - ***- acetylcholine***
  - ***- monoamines:***
    - *-Noradrenaline*
    - *-adrenaline*
    - *-dopamine*
    - *-serotonin , histamine*
  - ***- amino acids:*** *glutamate -aspartate- GABA - glycine*

- ***Other neurotransmitters:*** *represented above all by the peptides.*
  the case of nitrogen monoxide (NO or oxide nitric)
  ATP adénosine

VI/ THE NEUROMUSCULAR JUNCTION (NMJ)

1/Definition:

the NMJ designates the ***contact zone*** *between the* ***nerve ending*** *(axon of the motor neuronα)*

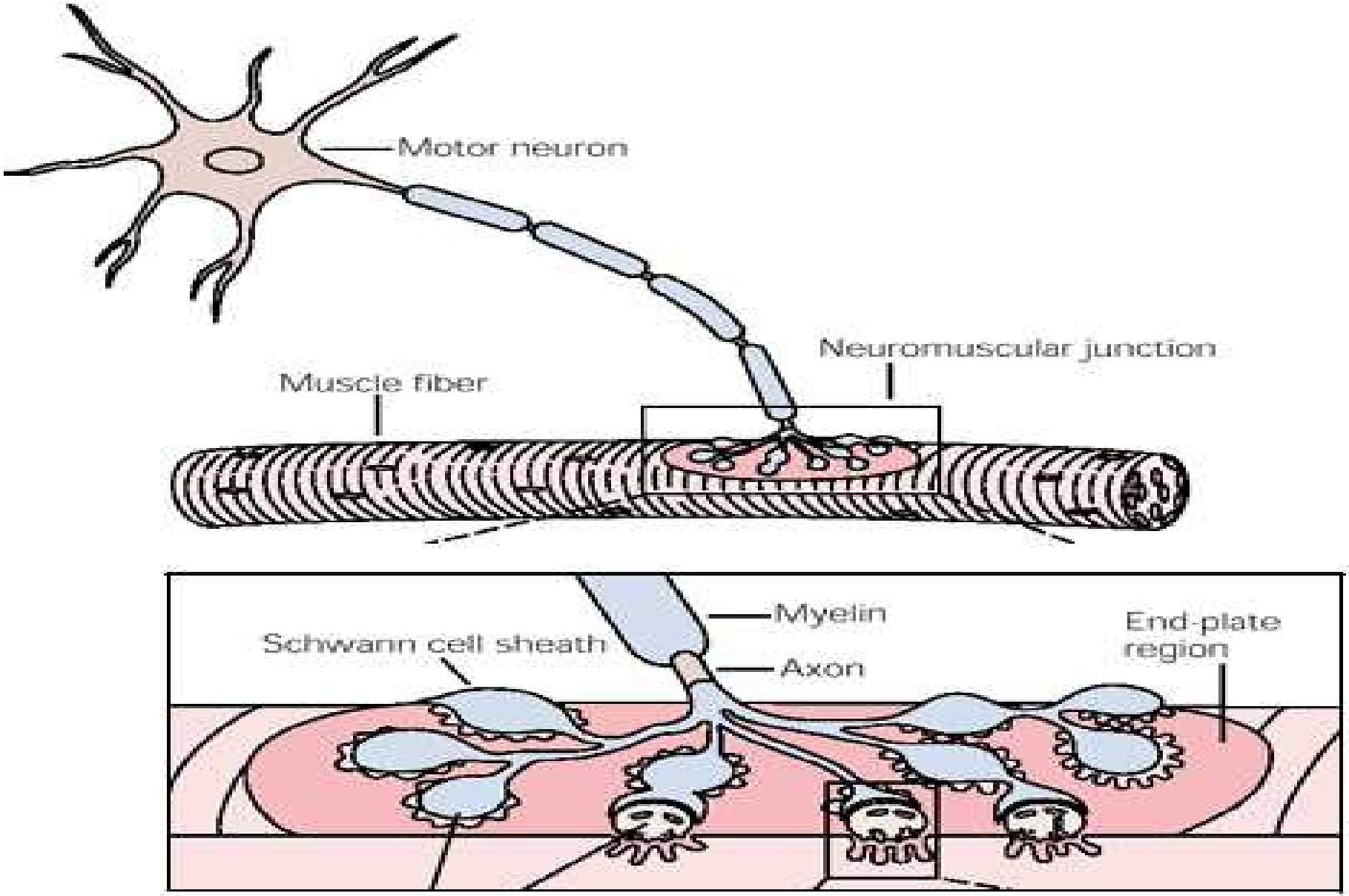
And ***muscle fiber*** *(striated muscle)*

skeletal) that it innervates, called ***plate motor****.*

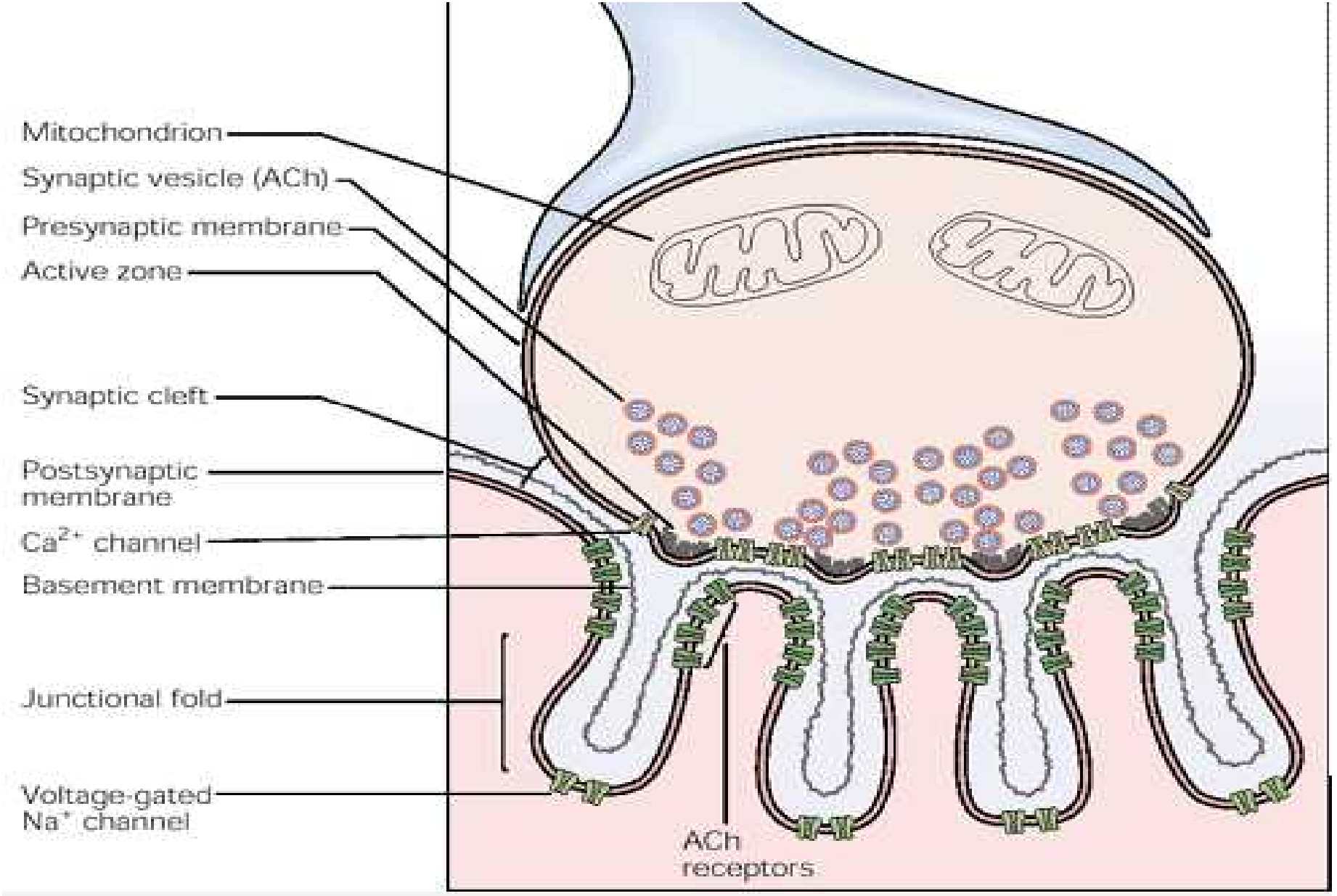
2/ STRUCTURE OF THE NMJ:

At the muscle level, the axon divides into branches, each of which innervates a single muscle fiber.
The terminal branches of the motor axon sink into a groove dug into the surface of the muscle membrane.
We can distinguish:
1-A pre-synaptic element, containing vesicles
synaptic receptors containing the neurotransmitter.
2-A synaptic cleft: 50 to 100nm wide.
3-A post-synaptic element: muscular membrane containing (at the level of the crests of the folds) the nicotinic acetylcholine receptors.

NMJ


Motor neuron
Neuromuscular junction
Muscle fiber
Myelin
Schwann cell sheath
Axon
End-plate region

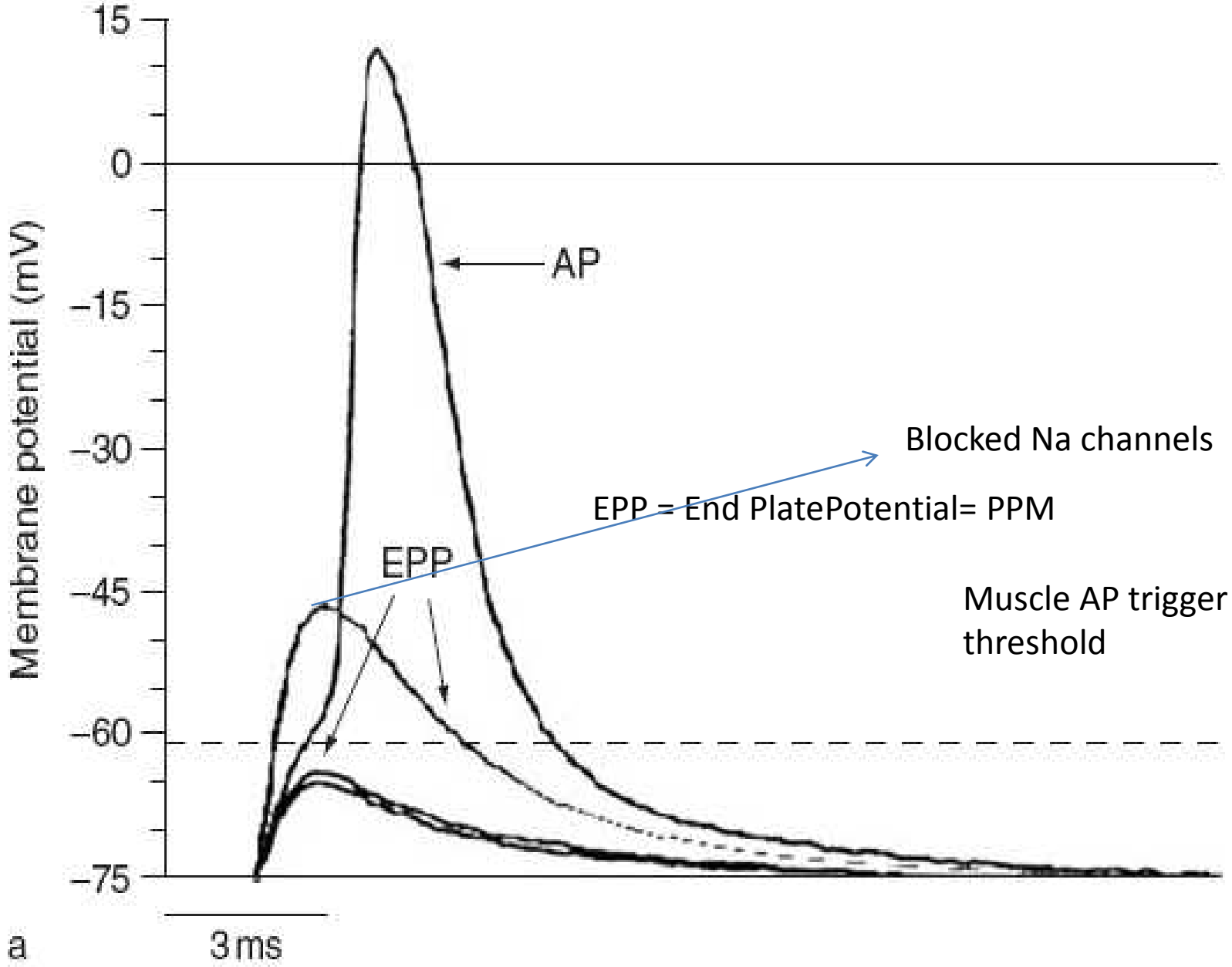
JNM


Mitochondrion
Synaptic vesicle (ACh)
Presynaptic membrane
Active zone
Synaptic cleft
Postsynaptic membrane
Ca^{2+} channel
Basement membrane
Junctional fold
Voltage-gated Na^{+} channel
ACh receptors

3/ELECTROPHYSIOLOGY:

a/ Endplate potential (EPP)

- *In the normal state, stimulation of the motor nerve always induces an action potential that propagates in the muscle membrane.*
- *After curarization* or blocking by TTX* tetrodotoxin, we can isolate a* ***local depolarization****: the endplate potential (EPP)*
- *EPP* ***is a local phenomenon****: reaches its maximum in 02 msec then decreases exponentially to zero in 20 msec.*


15
0
−15
−30
−45
−60
−75
Membrane potential (mV)
AP
EPP
3 ms
a
Blocked Na channels
EPP = End PlatePotential= PPM
Muscle AP trigger threshold

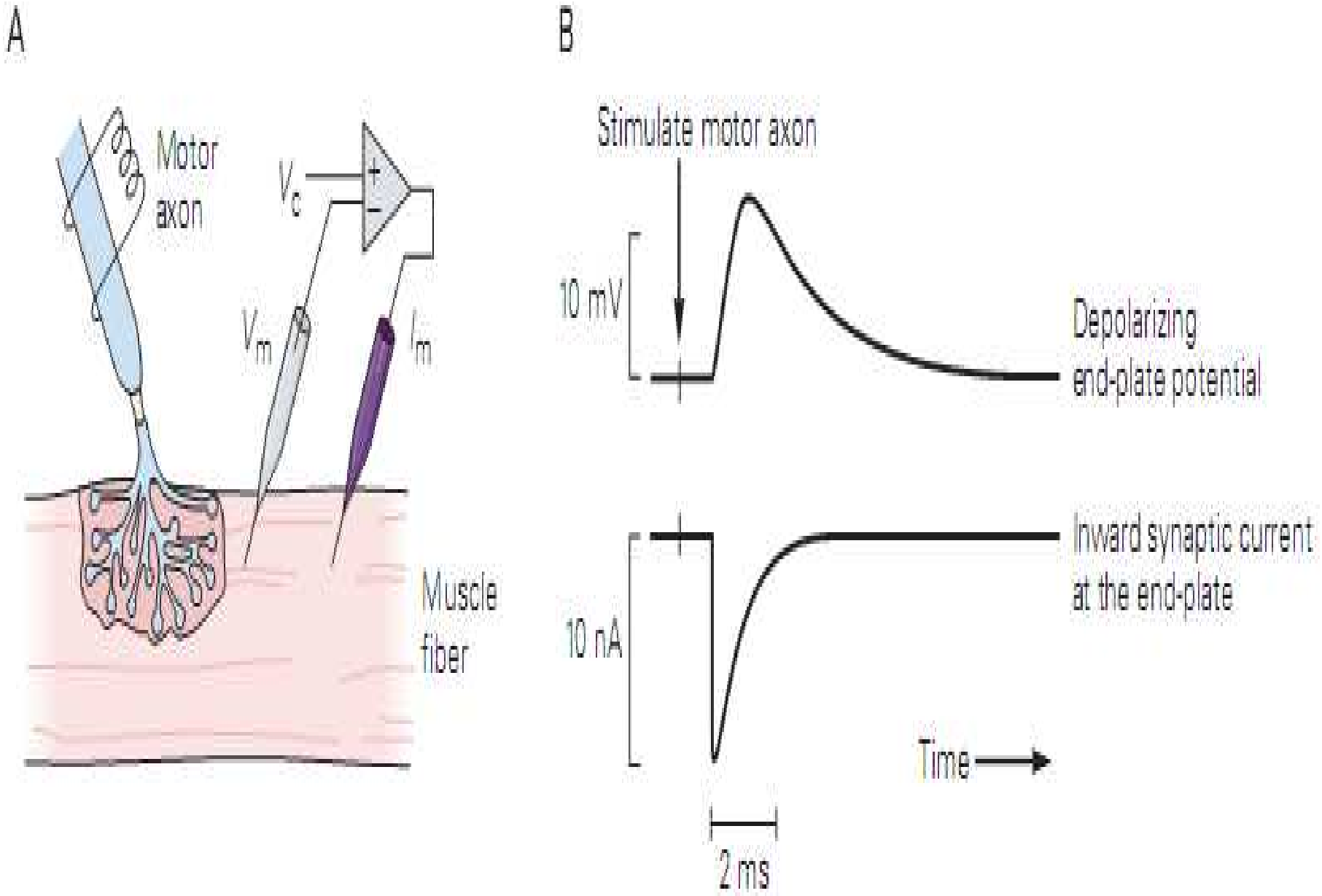
b/ IONIC MECHANISMS OF EPP

VOLTAGE CLAMP EXPERIMENT:

During the EPP, a decrease is observed membrane resistances with increase permeabilities and passage of ions.

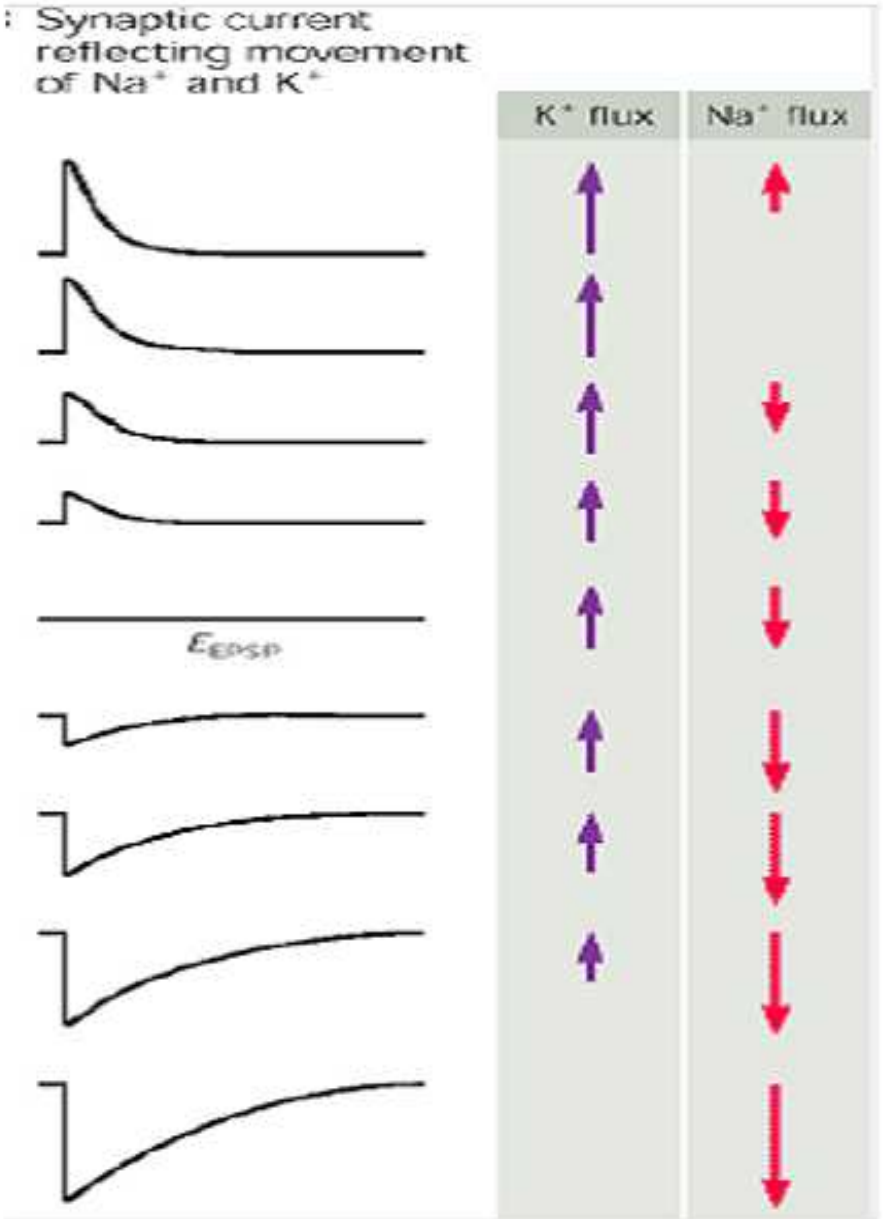
-The EPP is due to a simultaneous increase in ionic conductances for Na+ and K+ ions

-each ion tends towards the value of its potential equilibrium (E ion), the result is a value intermediate around -15mv.


A
Motor axon
V_c
+
−
V_m
I_m
Muscle fiber
B
Stimulate motor axon
10 mV
Depolarizing end-plate potential
10 nA
Inward synaptic current at the end-plate
Time
2 ms

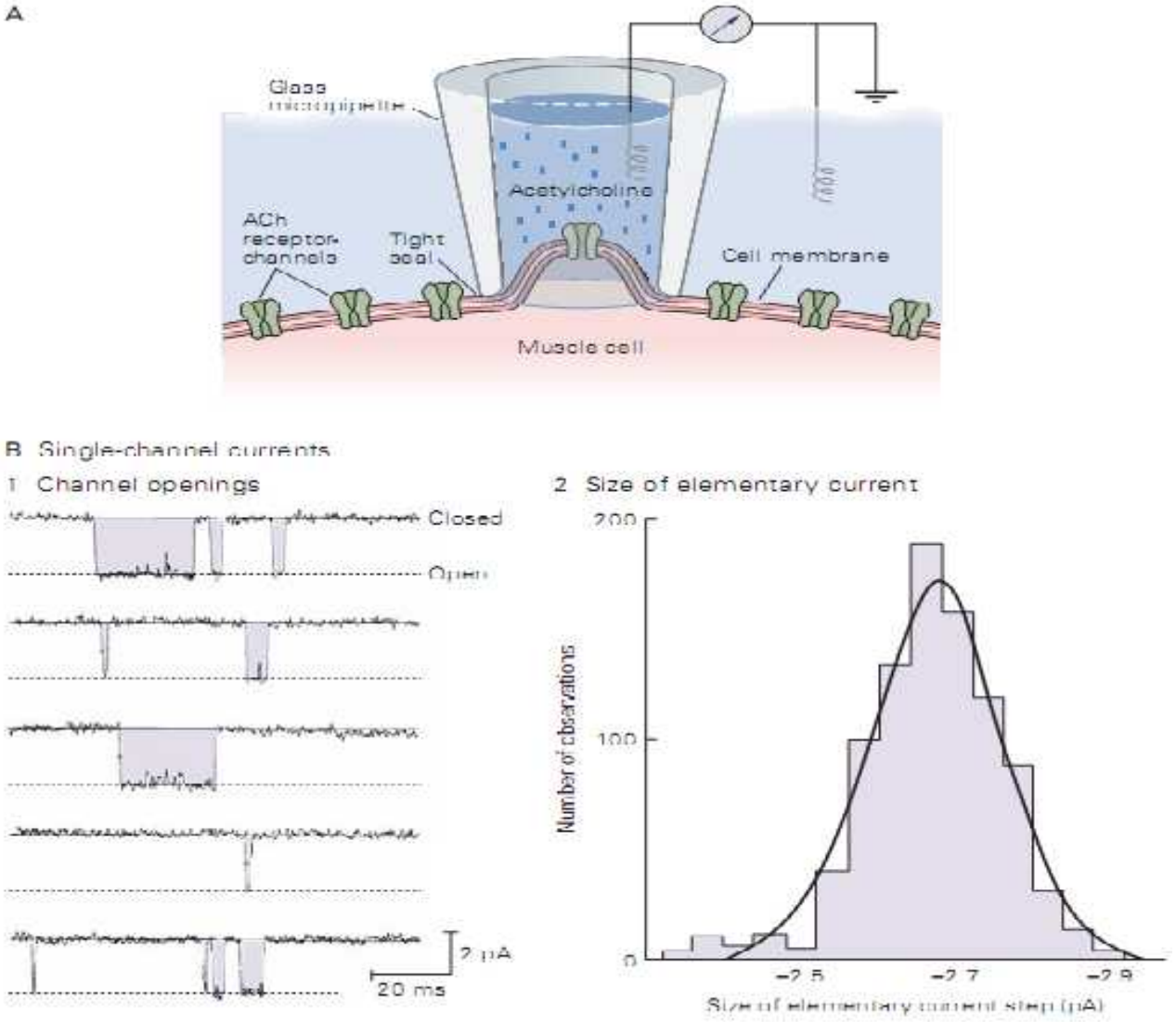
-Le PPM est du à une augmentation simultanée des conductances ioniques pour les ion Na+ et K+

-chaque ion tend vers la valeur de son potentiel d'équilibre (E ion), le résultat est une valeur intermédiaire autours de -15 mv.



The PATCH CLAMP technique:

allows to measure conductances and ionic current at the scale of a single channel.

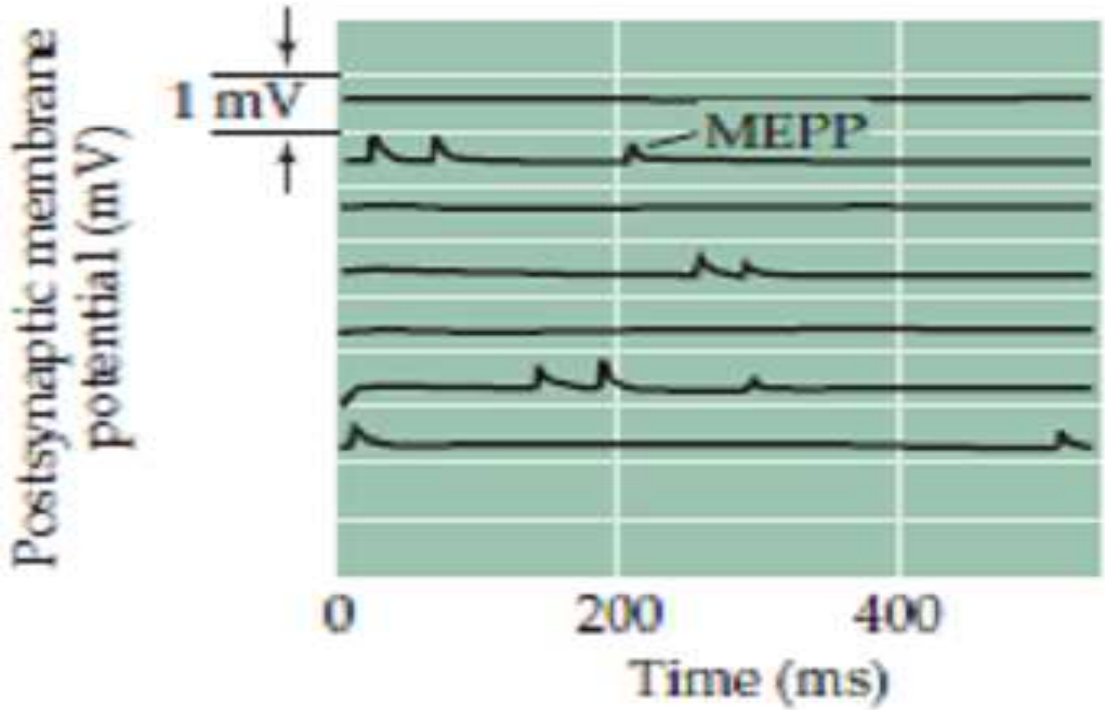


➢ *Miniature end-plate potential: (MEPP)*

Recorded without any stimulation the NMJ

is the seat of small potentials of 0.5mv amplitude

occurring spontaneously and randomly and would be the result of the release of small amounts of acetylcholine called quanta.



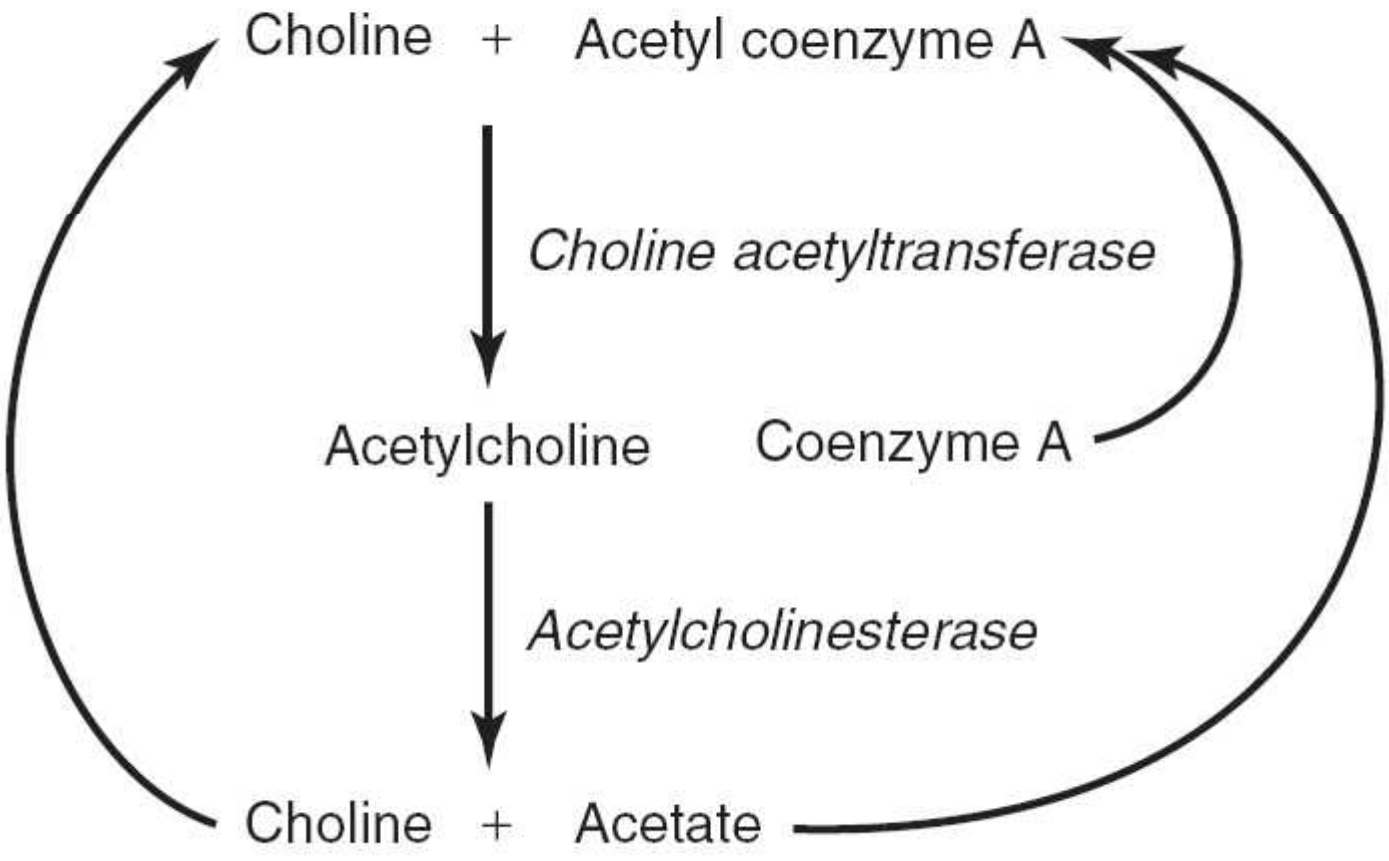
4/BIOCHEMICAL PHENOMENA

a/ Metabolism of acetylcholine:

- ***synthesis**: L'Ach is synthesized in the cytosol at the level of the nerve ending presynaptic from **acetylcoenzyme A** (which comes from the metabolism of glucose in mitochondrial level) and choline. This reaction is **catalyzed by choline acetyl transferase (CAT).***

Acetylcholine

- **Acetylcholine**

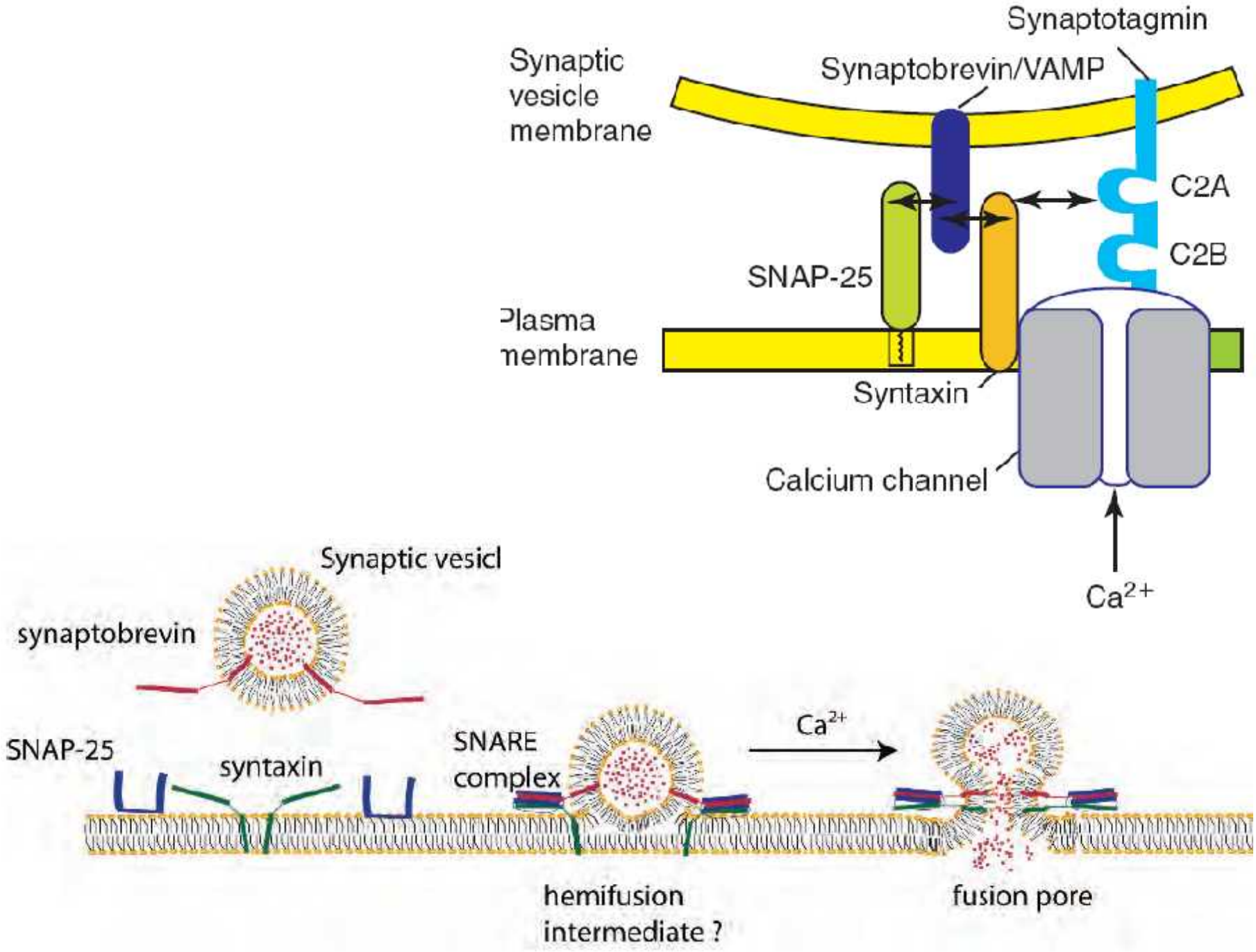


➢ ***Storage:*** *once the Ach synthesized is concentrated and stored in synaptic vesicles.*
In the synaptic button the vesicles are grouped into two pools, one reserve, the other located near the plasma membrane in the active zones ready for exocytosis

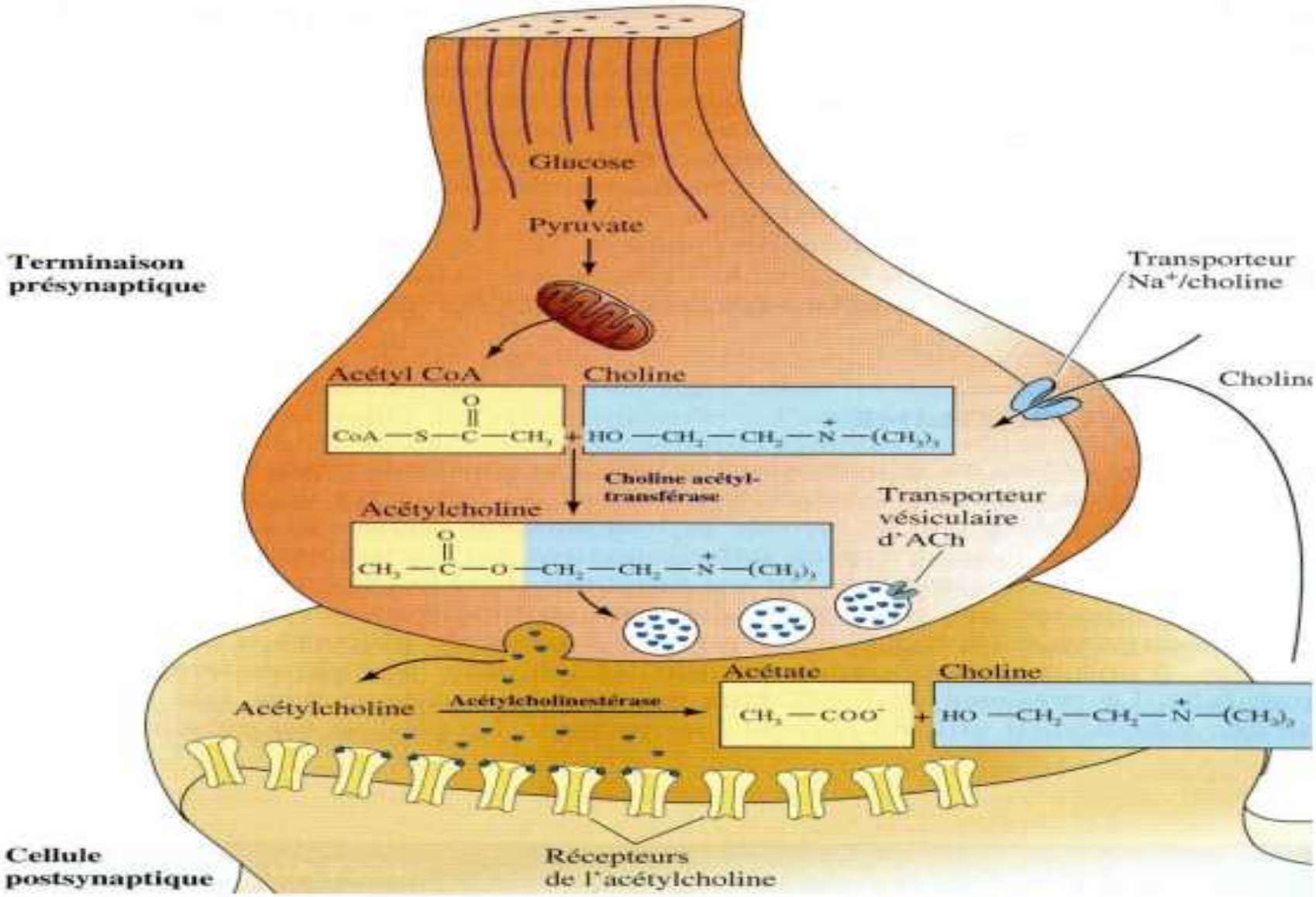
➢ ***L'exocytosis:*** *controlled by Ca++ which plays a role essential in the secretion of Ach*
The arrival of a PA at the termination level
Presynaptic induces a very rapid opening of the ***voltage-gated Ca++ channels***
Calcium entry will trigger the phenomena of fusion and exocytosis.

- ***Molecular mechanism of exocytosis:***

The increase inconcentration of Ca++ *in the induced presynaptic terminal* ***activation*** *of some* ***proteins*** *notably* ***there synaptotagmin*** *(which is a vesicular protein) with the participation of other proteins such* ***the SNARE complex (****there synaptobrevin vesicular, SNAP25 and the syntaxin of the plasma membrane) thus causing the docking of synaptic vesicles filled with Ach, the fusion of the vesicular and plasma membranes and release by exocytosis of the Ach in the synaptic space.*


Synaptotagmin
Synaptic vesicle membrane
Synaptobrevin/VAMP
C2A
C2B
SNAP-25
Plasma membrane
Syntaxin
Calcium channel
Ca2+
Synaptic vesicl
synaptobrevin
SNAP-25
syntaxin
SNARE complex
Ca2+
hemifusion intermediate ?
fusion pore

- ***Degradation:*** *once exercised its action on the postsynaptic element,* ***L'Ach is quickly degraded by the enzyme acetylcholinesterase*** *in choline and acetic acid. Choline will be recaptured through the presynaptic terminal thanks to active transporters.*



ACETYLCHOLINE CYCLE

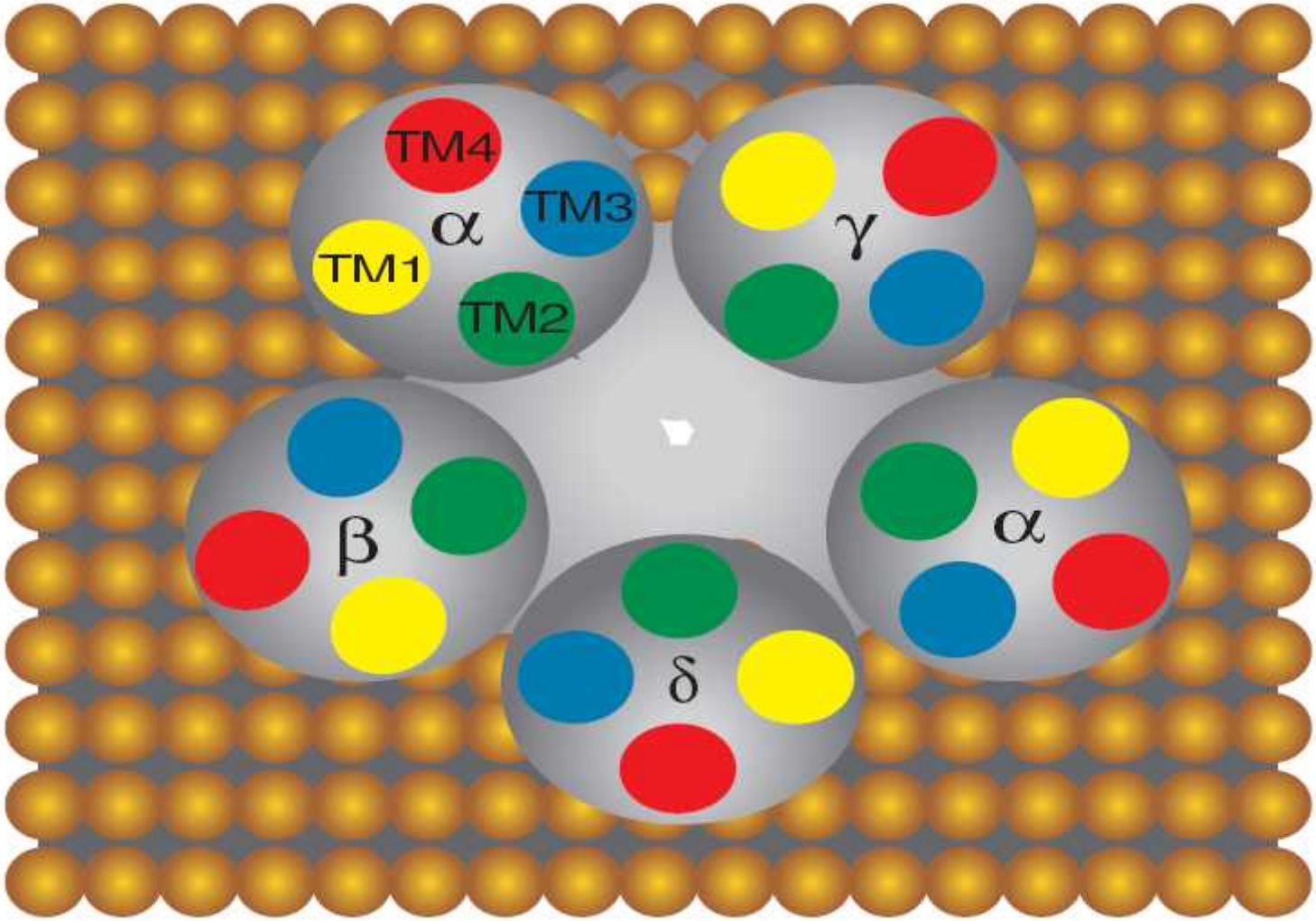
b/ Structure of the nicotinic ACH receptor:

It is a glycoprotein formed of 5 subunits: 2α,
β, γ, & and constituting the walls of a central pore.

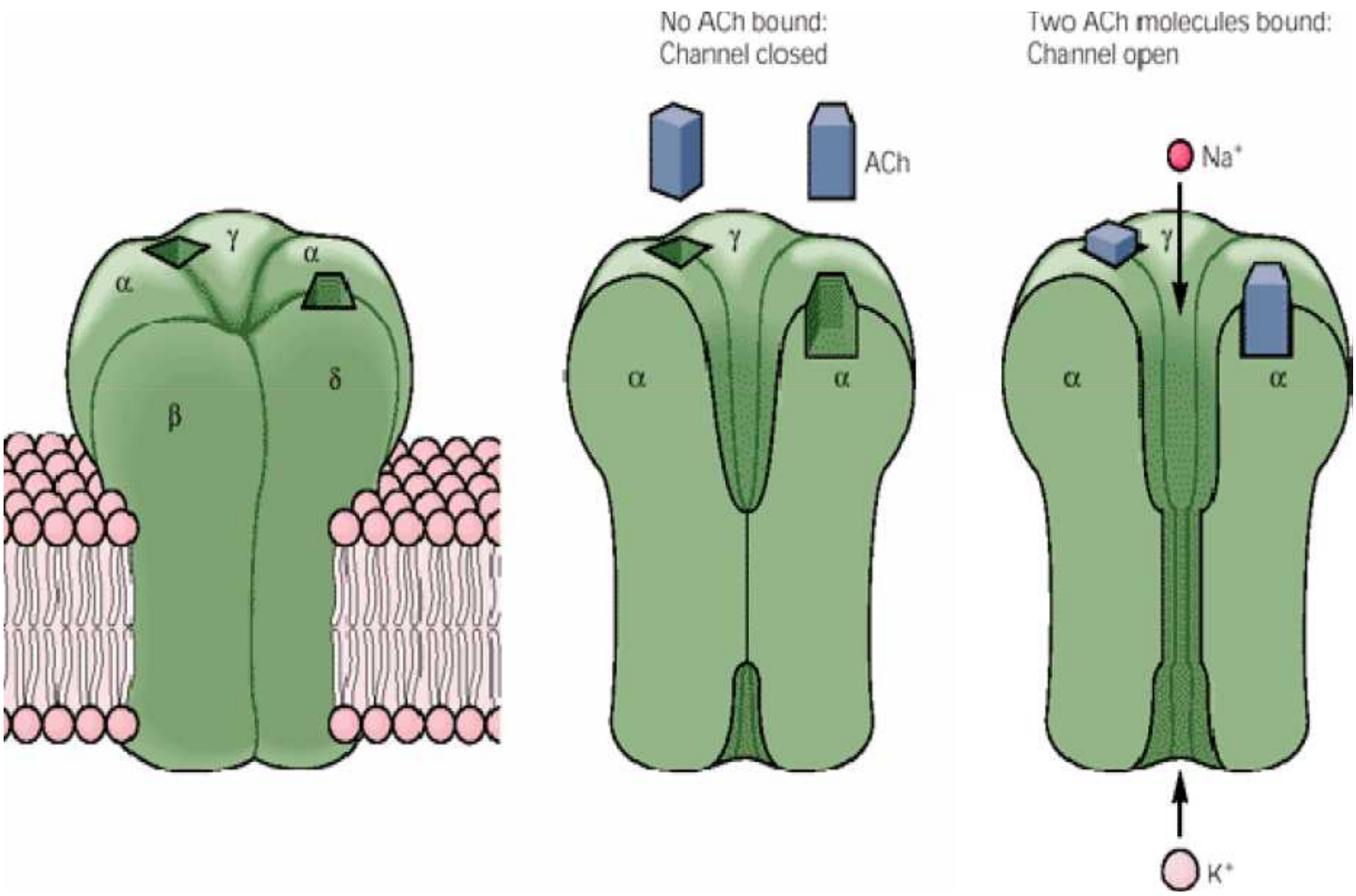
How it works:

The attachment of two ACH molecules to the receptor
nicotine causes a simultaneous increase,
rapid and transient membrane permeability
to cations (Na+ and K+). This increase in the
permeability is secondary to a change in
conformation of the nicotinic channel which changes
from the state "closed" to the "open" state.

Muscle nicotinic Ach receptor : upper view: note the 5 subunits 2 alpha subunits and 1 beta unit, 1 gamma unit and 1 delta unit. For each subunit the 4 transmembrane domains are represented



ACTIVATION OF POSTSYNAPTIC CHOLINERGIC RECEPTORS OF THE NMJ



6/ PHARMACOLOGY OF THE NEUROMUSCULAR JUNCTION:

Neuromuscular transmission can be blocked in three ways:

1-Inhibition of ACH synthesis.

2-Inhibition of release.

3-By interfering with the post-synaptic action of ACH.

1-Agents inhibiting ACH synthesis:

Hemicholinium

Vesamicol

2-Agents blocking the release of ACH:

-Calcium entry inhibitors: Mg++, antibiotics aminoglycosides (streptomycin, neomycin), calcium channel blockers.

-Neurotoxins: botulinum toxin, β bengarotoxin.

3-Agents acting post-synaptically:

3-1-Non-depolarizing blocking agents:

They are competitive antagonists of ACH at the level of the motor plate We distinguish:

-Reversible competitive antagonists: curare.

-Irreversible competitive antagonists: αbungarotoxin.

3-2-Depolarizing blocking agents:

These are ACH agonists, examples: decamethonium, suxamethonium.

7/ Clinical application

Postsynaptic disorders

a/ Myasthenia gravis: it is a condition autoimmune characterized by the presence of antibodies to the receptors to the Ach

b/ myasthenic congenital Syndromes

c/ Toxic damage postsynaptic

Presynaptic disorders

a/ Lambert Eaton syndrome : it is an affection Autoimmune characterized by the presence of Antibodies calcium channel blockers

b/ Botulism: the toxin blocks the release of Ach

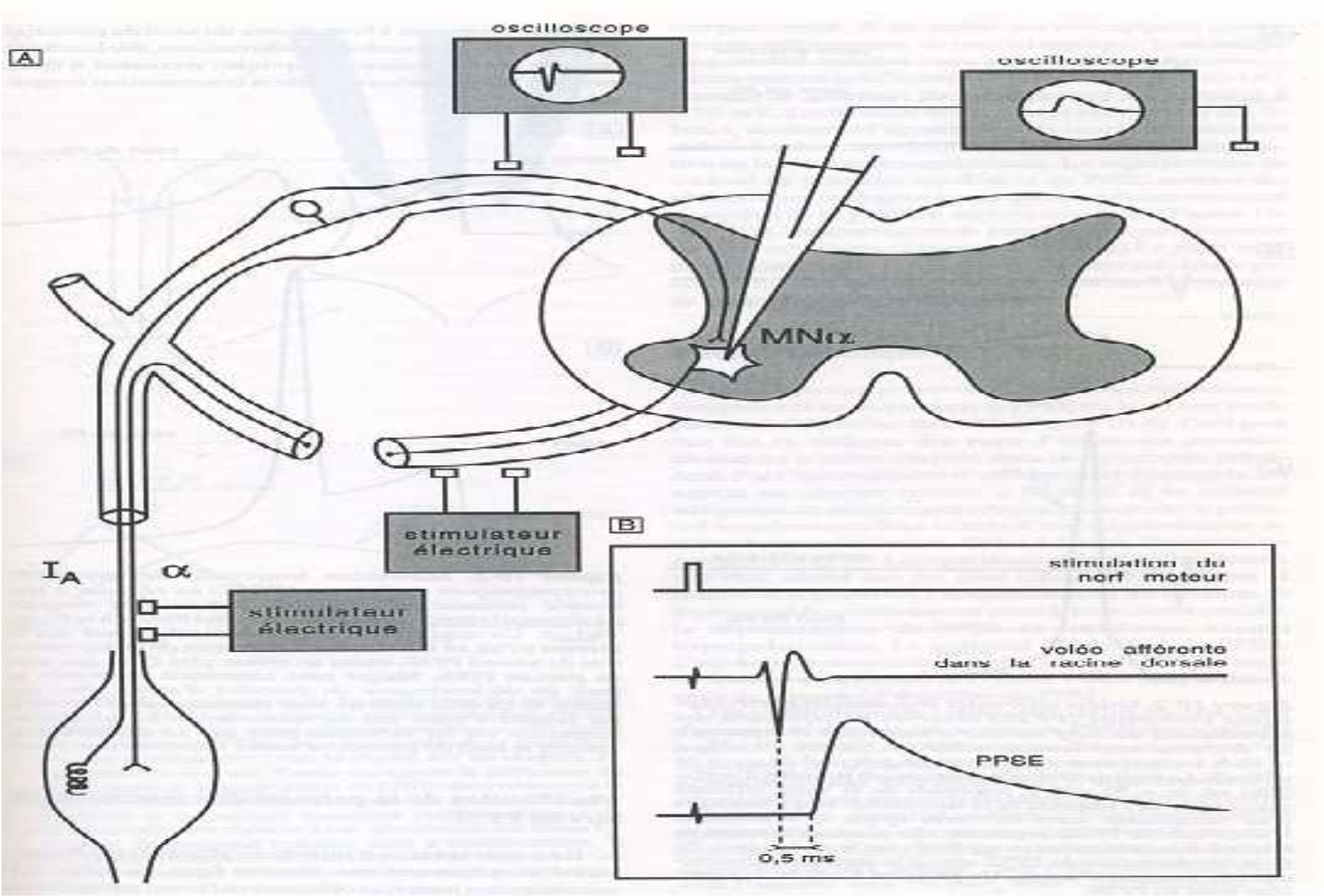
VII/ neuroneuronal Synapses:

The neuroneuronal synapses is a contact zone between 02 neurons.

1/ Electrophysiology:

A/ The excitatory postsynaptic potential EPSP:

a/ Highlighting the EPSP: diagram



Post-synaptic electrical phenomena EPSP

b/ Characteristics of EPSP:

EPSP is a local depolarization of the membrane postsynaptic secondary to the fixation of the excitatory neurotransmitter on receptors specific.

EPSPs establish themselves quickly and disappear Gradually.

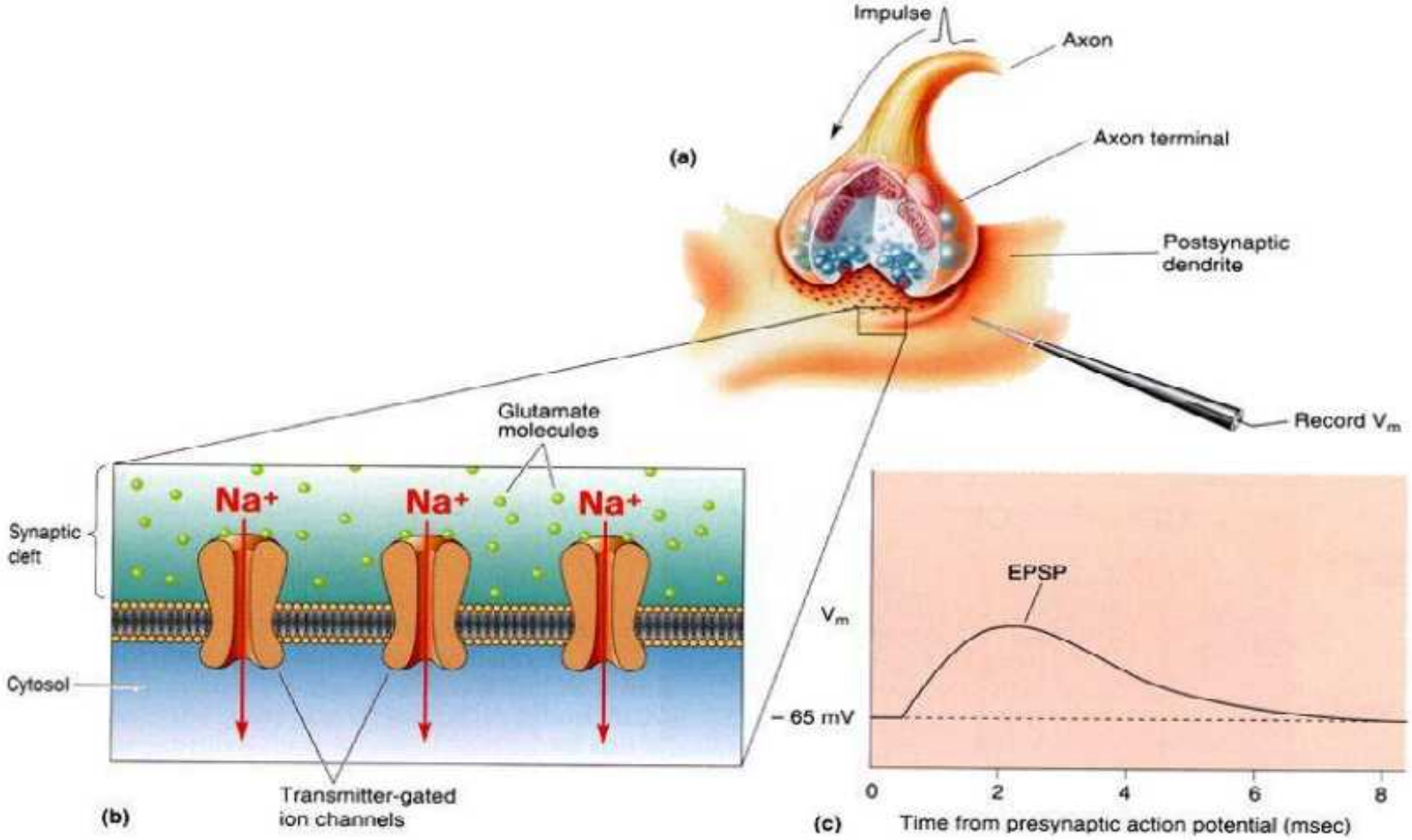
The EPSP are summable, gradable (the amplitude of the PPSE depends on the amount of the neurotransmitter released) decremental (the amplitude of the EPSP decreases with distance)

The EPSP does not obey the all or nothing law and does not does not have a refractory period.

c/ Ionic mechanism of EPSP:

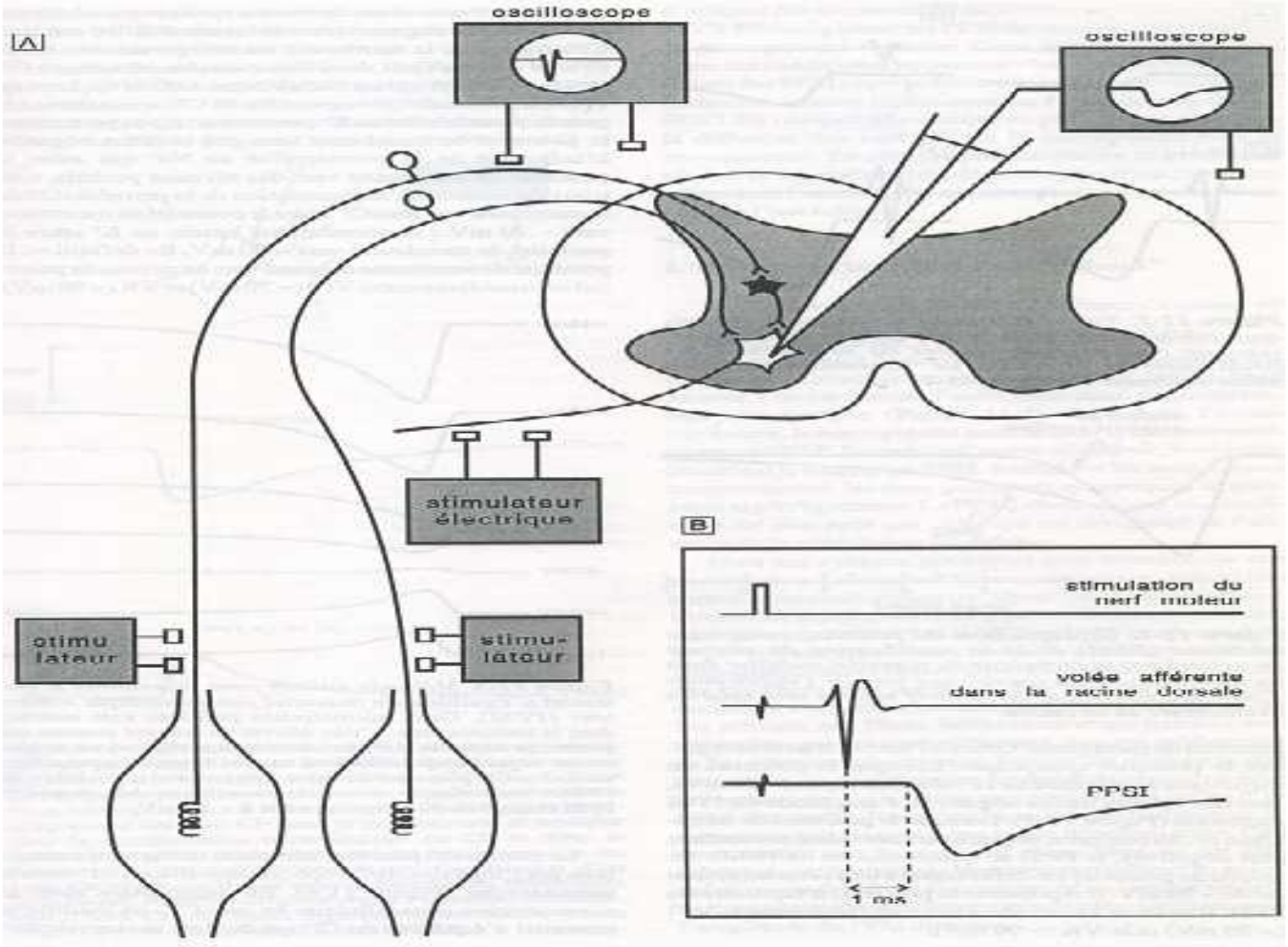
The EPSP results from the simultaneous increase in

The membrane permeability gNa+,gK+



B/ The inhibitory postsynaptic potential IPSP:

a/ Highlighting the IPSP:



***b/ characteristics of IPSP*:**

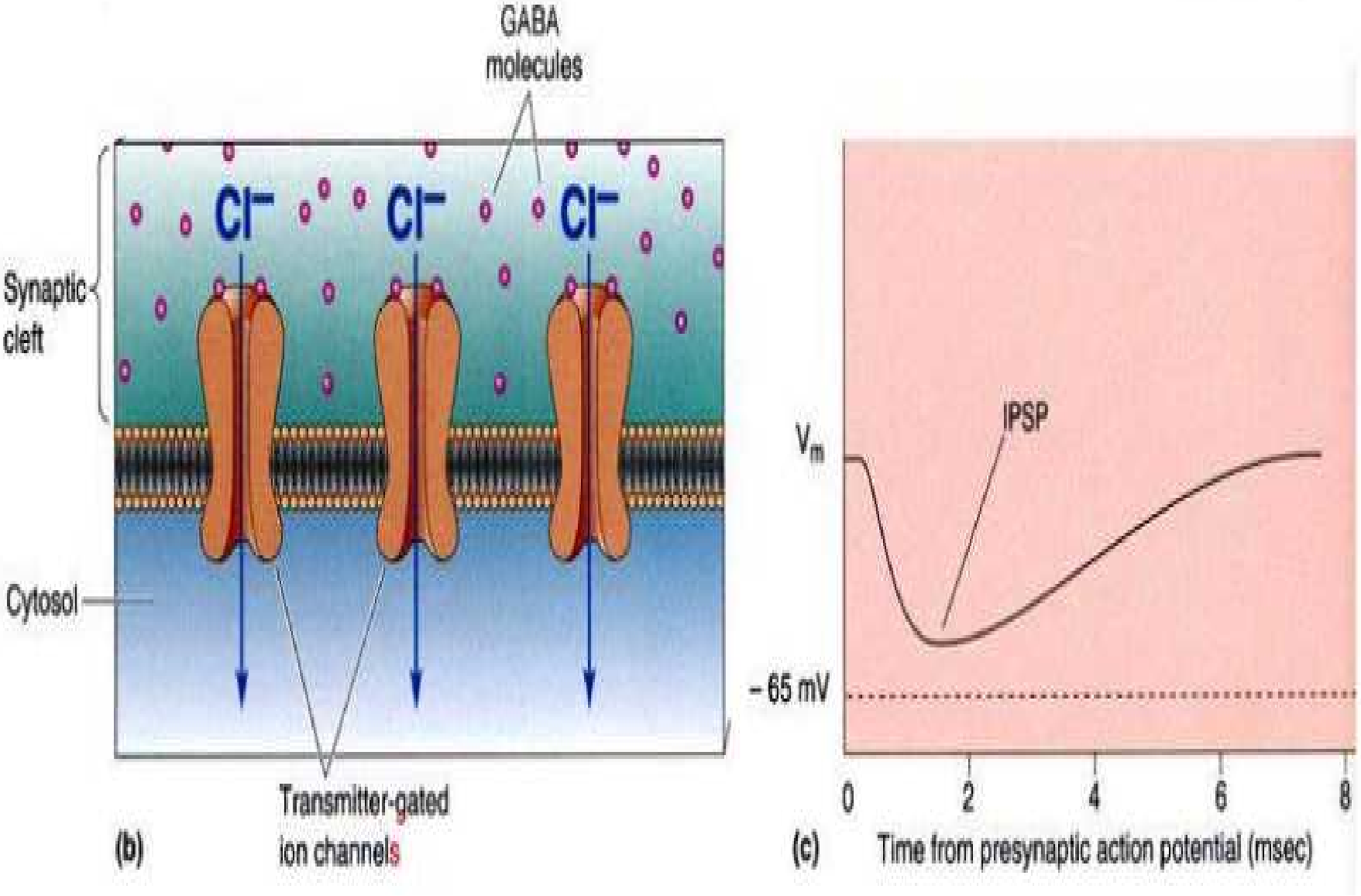
It is the mirror image of the EPSP the IPSP is a membrane hyperpolarization postsynaptic secondary to neurotransmitter binding inhibitor on specific receptors.

c/ Ionic mechanism of IPSP: *the IPSP is a membrane hyperpolarization which results from:*

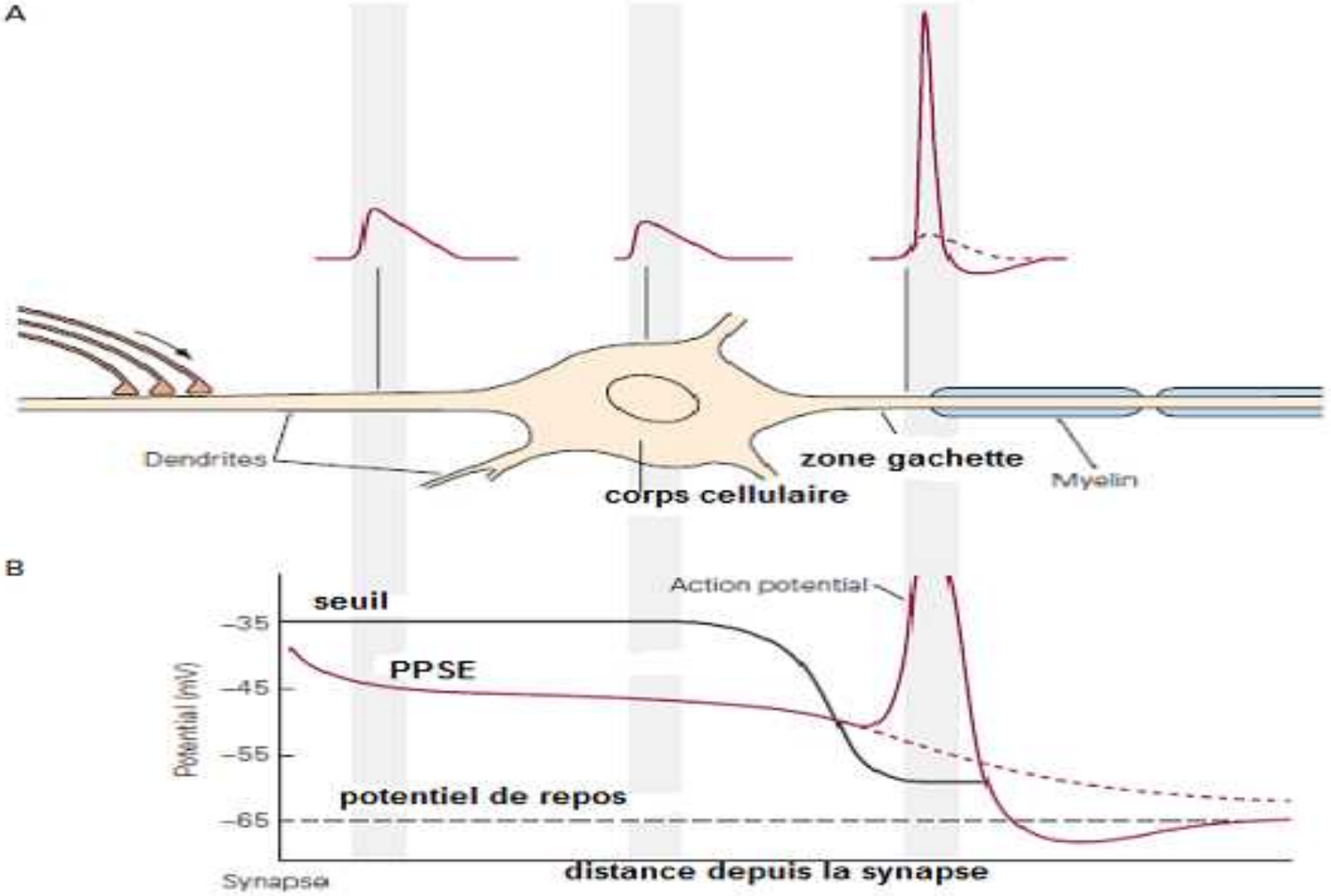
From a flow of Cl- anion entering the cell

From an outgoing flow of K+ cations.

Ionic mechanism of IPSP



2/ Synaptic integration

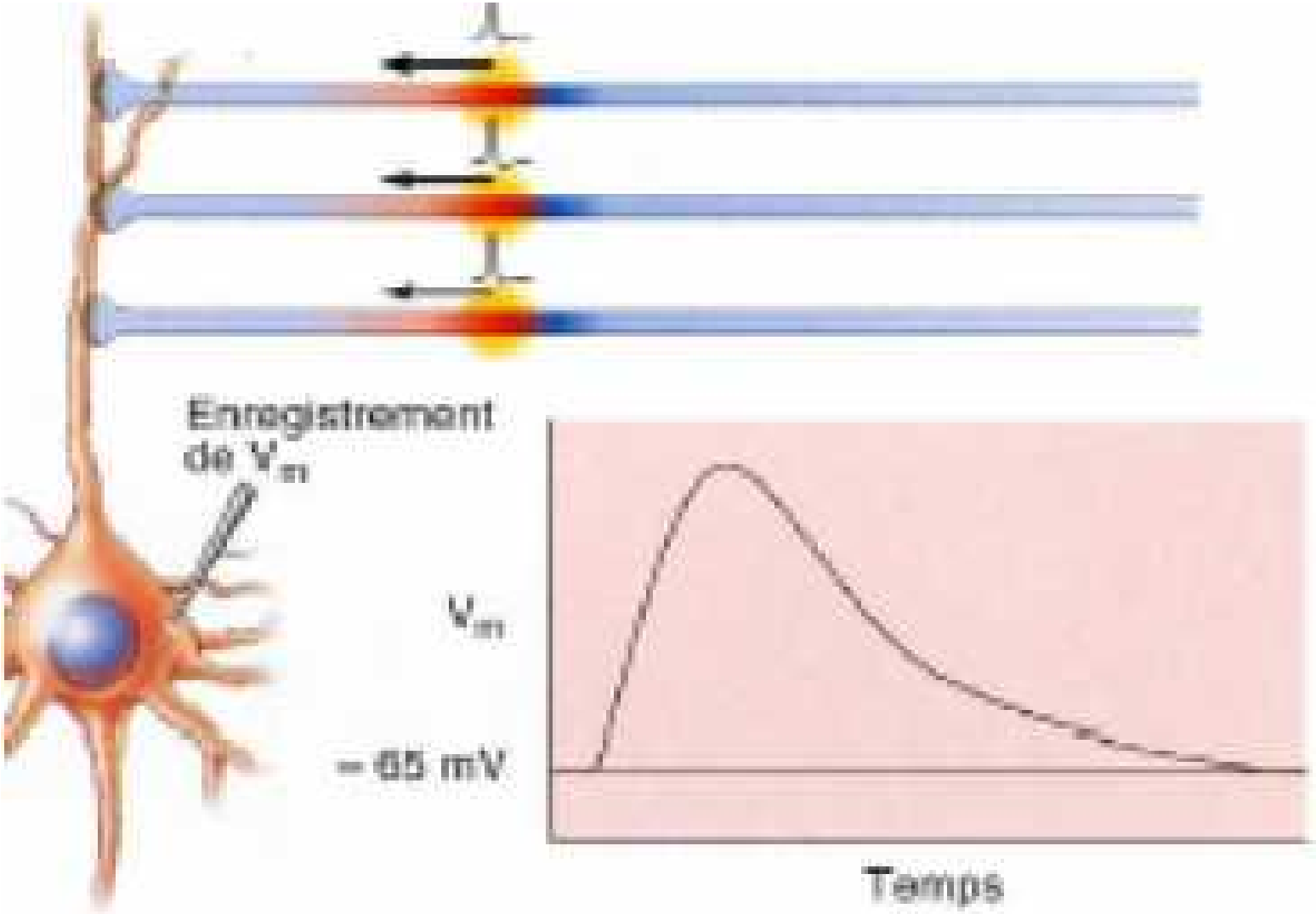


-The initial segment (trigger zone) is the site of birth of the neuronal action potential

-At this level, membrane excitability is maximal (abundance of voltage-dependent Na+ channels)

-in figure A, the sum of the EPSP when it exceeds the threshold and therefore can induce a AP at the level of the initial segment (AP: action potential)

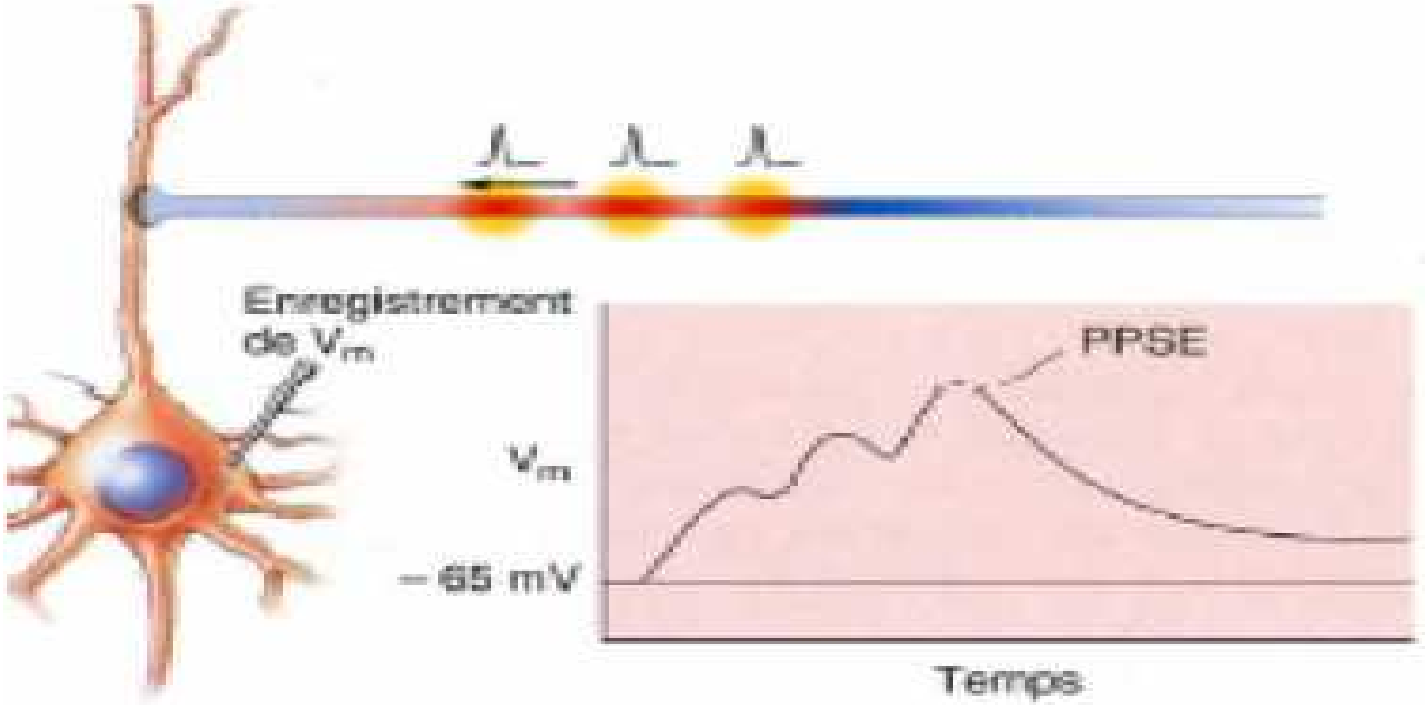
a/ Spatial summation: *the same neuron Presynaptic stimulates the post-synaptic neuron repeatedly in a short period of time the resulting depolarization of all of these EPSP leads to the appearance of a AP.*



Sommation spatiale

b/ Temporal summation

Several neurons stimulate the same neuron Postsynaptic all of these EPSPs lead to the appearance of a AP



Sommation temporelle

3/ presynaptic Inhibition :

